

Manuel Batalha dos Santos

 manuel.batalha.santos@gmail.com |  @manel1874 |  @manel1874 |  @manel1874 |  xUARChoAAAAJ

WORK EXPERIENCE

LEAD CRYPTOGRAPHY RESEARCHER | Remote

MULTIVM LABS | April 2026 – Now

- Research quantum attacks on ECDSA (ECDSA.Fail) and help building Potomac quantum cryptanalysis team.
- Implement threshold post-quantum signatures and design TEE co-signer architectures for hybrid signing.
- Conduct quantum-readiness audits for Tether, Canton and OG.

CRYPTOGRAPHY RESEARCH ENGINEER | Remote

TECTONIC LABS | October 2025 – March 2026

- Implemented threshold ML-DSA post-quantum protocol ().
- Designed and implemented () hybrid hierarchical deterministic wallet for post-quantum signatures ( Blog).
- Maintainer of Tectonic's cryptographic library providing post-quantum schemes ( Bedrock).
- Conducted internal security auditing to PQWallet and patched 4 high- and 6 medium-risk findings.

CRYPTOGRAPHY ENGINEER | Remote, Lisbon

NILLION LABS | May 2023 – August 2025

- Research zkTLS integration with secure computation technologies (TEE, MPC and FHE).  Part 1, 2, 3.
- Translated cryptographic protocols into production-ready implementations ( nilVM) and contributed to Nillion's MPC programming language (Nada and Nada-numpy).

CRYPTOGRAPHY RESEARCHER | Lisbon, Portugal

TELECOMMUNICATIONS INSTITUTE | Oct 2019 – May 2023

- Developed a secure genome analysis system (QuPPA) based on custom forks of MP-SPDZ and Libsapi.

DATA SCIENTIST | Lisbon, Portugal

TEKEVER | Nov 2018 – Sep 2019

- Built a recommendation system for Portugal's Trade Agency.

QUANTUM CRYPTOGRAPHY RESEARCHER | Portugal

TEKEVER | Dec 2017 – Nov 2018

- Raised €609 544 of funding for the Q.DOT project.

ADDITIONAL SKILLS

PROGRAMMING

PROFICIENT: Rust • Python • C/C++
Neo4j • Docker • Git

FAMILIAR: JavaScript, HTML, SQL

LANGUAGES

Fluent English
Native Portuguese
Basic Spanish

AWARDS

NEW TALENTS IN MATHEMATICS

CALOUSTE GULBENKIAN FOUNDATION | Oct 2015 – Jul 2016

Research scholarship assigned to 20 undergraduate Portuguese students (Research topic: Dynamical Systems and Ergodic Theory).

FCT SCHOLARSHIP

FOUNDATION FOR SCIENCE AND TECHNOLOGY (FCT)

PhD scholarship granted to 10 Mathematics candidates nationwide.

EDUCATION

PHD IN INFORMATION SECURITY

TÉCNICO, UNIVERSITY OF LISBON

Distinction | Grad. Jan 2025

 Secure Multiparty Computation assisted with quantum primitives

MSC IN APPLIED MATHEMATICS

IMPERIAL COLLEGE LONDON

First-Class Honours | Grad. Oct 2017

 Topological Band Systems and Finite Size Effects

BSC IN APPLIED MATHEMATICS AND COMPUTATION

TÉCNICO, UNIVERSITY OF LISBON

17/20 | Grad. Jun 2016

SELECTED PROJECTS

 OTLS: Integrated RELIC lib for pairing-based elliptic curve zkTLS protocols.

 TFHE-rs: Implemented ZK proofs for LWE encryption.

 emp-zk: Built ZKP circuits for LWE-based encryption.

 LFDT-Lockness: Contributed to the CGGMP ECDSA threshold signing protocol upgrade.

 nilVM: Contributed to implementing MPC cryptographic protocols and integrated threshold signatures protocols.

 Curl: Implemented LUT-based protocols for non-linear functions and implemented truncation protocol.

 thmldsa-rs: Implemented Mithril (threshold post-quantum ML-DSA) protocol.

 nilRAG: Privacy-preserving RAG with MPC and TEE.

 Blind Lawyer: Web app to analyze legal documents with privacy. Winner of the Nillion internal hackathon.

SELECTED PUBLICATIONS

Research

 ECDSA.Fail: Open Autoresearch for Optimizing Elliptic-Curve Point Addition in Shor's Algorithm, arXiv, 2026.

 Post-Quantum Blockchains with Agility in Mind, MAGiCS'26.

 TLShare: Private Authenticated MPC and FHE Inputs Over TLS, ePrint Archive, 2025 .

 Fission: Distributed Privacy-Preserving Large Language Model Inference, IEEE Communications and Network Security (CNS), 2025.

 Curl: Private LLMs through Wavelet-Encoded Look-Up Tables, Camlis, 2024 .

 Quantum Universally Composable Oblivious Linear Evaluation, Quantum, 2024.

 PECO: methods to enhance the privacy of DECO protocol, ePrint Archive, 2022.

 Quantum Oblivious Transfer: A Short Review, Entropy, 2022.

Blogposts

 Exploring Signature-Free Post-Quantum RLPx Handshake

 Hybrid Hierarchical Deterministic Wallets

 Evolving zkTLS: Privacy-Preserving Computation from Decentralized Oracles. Part 1, 2, 3.

 Privacy-Preserving Computational Biology using Yao protocol.